International Institute of Information Technology, Bhubaneswar

Department of Computer Science and Engineering

Object Oriented Programming Laboratory

Lab 2 – C++ Programming Assignment (Group B)

Date: 07.08.2026 **Semester:** B.Tech 3rd Semester **Section:** CSE B1

Instructions

- Write all programs in **C++**.
- Use **classes and objects** in every program.
- Implement the required functionality using **member functions**.
- Display the output in a neat and readable format.
- Use appropriate data types and meaningful variable names.

1. **Car Information System**
   Create a class named `Car` to store the details of a car. The class should contain the following data members:

   - Car Number
   - Brand Name
   - Model Year

   Write suitable member functions to:

   1. Accept the car details from the user.
   2. Display the entered details in a formatted manner.

   Hint: *Use separate functions for input and display.*

2. **Square Calculator**
   Create a class named `Square` that stores the side of a square. Write member functions to:

   1. Read the side length.
   2. Calculate the area.
   3. Calculate the perimeter.
   4. Display the results.

   Hint: *Area* = *Side* $\times$ *Side, and Perimeter* = 4$\times$ *Side.*

3. **Temperature Converter**
   Create a class named `Temperature`. Write member functions to:

   1. Accept temperature in Celsius.
   2. Convert it into Fahrenheit.
   3. Display both temperatures.

   Hint: *Use the formula:* $F = \frac{9}{5}C + 32$

4. **Hotel Room Booking**
   Design a class named `HotelRoom`. The class should contain:

   - Room Number
   - Guest Name
   - Number of Days Stayed
   - Cost Per Day

   Write member functions to:

   1. Accept booking details.
   2. Calculate the total room rent.
   3. Display the booking details.

   Hint: *Total Rent = Number of Days Stayed* $\times$ *Cost Per Day.*

5. **Mobile Recharge System**
   Create a class named `MobileRecharge`. The class should contain:

   - Mobile Number
   - Customer Name
   - Current Balance

   Write member functions to:

   1. Accept customer details.
   2. Recharge the balance.
   3. Deduct the balance after selecting a recharge plan.
   4. Display the updated balance.

   Hint: *Ensure that the recharge amount does not exceed the available balance.*

6. **Time Addition**
   Create a class named `Time` with two data members: **Hours** and **Minutes**. Write member functions to:

   1. Input two time values.
   2. Add the two time values.
   3. Display the resulting time.

   Hint: *Convert every 60 minutes into 1 hour. Example: 2 hr 45 min + 1 hr 30 min = 4 hr 15 min.*

7. **Movie Ticket Booking**
   Create a class named `MovieTicket`. The class should contain:

   - Movie Name
   - Ticket Price
   - Number of Tickets

   Write member functions to:

   1. Accept booking details.
   2. Calculate the total ticket cost.
   3. Display the booking summary.

   Hint: *Total Cost = Ticket Price $\times$ Number of Tickets.*

8. **Hostel Fee Management**
   Create a class named `HostelFee`. The class should contain:

   - Student Name
   - Hostel ID
   - Monthly Fee
   - Number of Months

   Write member functions to:

   1. Accept student details.
   2. Calculate the total hostel fee.
   3. Apply a late fine of ₹500 if the payment is delayed.
   4. Display the final amount payable.

   Hint: *Use an additional variable to indicate whether the payment is delayed.*

9. **Cricket Score Analyzer**
Create a class named `CricketPlayer`. The class should contain:

   - Player Name
   - Matches Played
   - Total Runs Scored

   Write member functions to:

   1. Accept player details.
   2. Calculate the batting average.
   3. Classify the player's performance using the following criteria and display the complete player report:

Batting Average	Performance
50 and above	Excellent
35–49.99	Good
20–34.99	Average
Below 20	Poor

   Hint: *Batting Average = Total Runs / Matches Played.*

10. **Water Bill Calculator**
Create a class named `WaterBill`. The class should contain:

   - Consumer Number
   - Consumer Name
   - Water Consumption (in litres)

   Write member functions to accept the consumer details, calculate the water bill using the following slab rates, and display the complete bill.

Water Consumption	Rate
First 500 litres	₹2 per litre
Next 500 litres	₹3 per litre
Above 1000 litres	₹5 per litre

   Hint: *Calculate the bill slab-wise instead of multiplying all litres by a single rate. Example for 1200 litres: (500 × ₹2) + (500 × ₹3) + (200 × ₹5) = ₹3500.*